

The Yang–Mills Mass Gap Reinterpreted Through Emergent Spacetime Geometry

Thomas Birnie

Abstract

We reinterpret the Yang–Mills mass gap phenomenon using the operator-theoretic framework of emergent spacetime. Bulk decay and boundary field coherence are not separate from spacetime itself—they generate its dimensionality. By aligning the energy transfer dynamics of bulk–boundary field simulations with the embedding of non-commutative (NC) geometry into classical manifolds, we identify the mass gap as a spectral residue of emergent classicality from quantum operator space.

1 Introduction

The Clay Institute’s Yang–Mills Gap problem posits the existence of a finite, non-zero lowest energy state (the “mass gap”) in four-dimensional $SU(N)$ Yang–Mills theory. Previous topological approaches often rely on the discretization of instanton sectors or geometric quantization. Our approach reinterprets this gap as a consequence of the emergent nature of spacetime itself, where dimensionality, curvature, and field coherence arise from operator-based interpolation mechanisms.

2 Bulk as a Non-Commutative Region

We interpret the “bulk” field region, where gauge configurations decay or destabilize, as an intrinsically non-commutative domain. Field operators $\hat{A}_\mu$ evolve in imaginary time τ , and geometry is non-metric:

$$\lim_{\theta \rightarrow 0} [\hat{x}^\mu, \hat{x}^\nu] \neq 0 \tag{1}$$

Simulations show that bulk field energy decays monotonically, matching the expected dissipative behavior of operator domains without classical spacetime constraints.

3 Boundary as a Classical, Commutative Region

At the system boundary, we observe:

- Convergence of field amplitudes
- Persistent field coherence
- Coupled growth following energy leakage from the bulk

This corresponds to the classical limit of the operator framework, where

$$\lim_{\theta \rightarrow \infty} [\hat{x}^\mu, \hat{x}^\nu] \rightarrow 0 \Rightarrow x^\mu \in \mathbb{R}^4 \quad (2)$$

Here, classical Yang–Mills dynamics emerge from the underlying operator space.

4 Embedding Function as Bulk–Boundary Mediator

We define the embedding function $\mathcal{E}$ mapping from non-commutative manifold M_{NC} to classical manifold M_C :

$$\mathcal{E}(M_{NC}, T^{\mu\nu}, R^{\mu\nu}, \theta, D_f) = \int_{M_{NC}} K(x, y, \theta) F(y) dV_y \quad (3)$$

Where K is a kernel encoding coupling geometry, D_f is the interpolating Dirac operator, and θ is a deformation parameter controlling the non-commutative to commutative transition.

5 The Gap as Spectral Signature

The “gap” emerges as a restriction on allowed transitions between field states:

- In NC domain: transitions are unbounded but unstable
- At boundary: only discrete stable configurations are realized

Thus, the Yang–Mills gap becomes a spectral residue—the lowest eigenvalue of boundary field coherence under the $\mathcal{E}$ map.

6 Temporal Interpretation

Time is not intrinsic to either domain. It is regulated by the complexified time operator:

$$\Theta = T + i\tau \quad (4)$$

where T governs classical progression and τ encodes quantum decoherence. Bulk decay occurs dominantly along τ , while boundary stabilization tracks T , giving rise to the arrow of time.

7 Conclusion

By reframing Yang–Mills theory within the emergent spacetime operator framework, we replace fixed spacetime assumptions with a dynamic field-driven geometry. The mass gap becomes a necessity of dimensional emergence, not a secondary effect of field topology.